ONE DISTRICT ONE PRODUCT
UTTAR PRADESH

A GUIDE TO PRODUCT QUALITY STANDARDS

MUSICAL INSTRUMRNTS

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INDEX

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ABBREVATIONS

ACCC	Australian Competition and Consumer Commission
ANSI	American National Standards Institute
APLAC	Asia Pacific Laboratory Accreditation Co-operation
APHIS	Animal and Plant Health Inspection Service
AS	Australian Standards
ASTM	American Society for Testing and Materials
BICON	Bio Security Import Condition System
BIS	Bureau of Indian Standards
CE	Conformité Européenne
CoC	Certificate of Conformity
CPSC	Consumer Products Safety Commission
EC	Council Regulation (of the European Parliament and Council)
ECHA	European Chemicals Agency
EN	European Standards
EPA	Environment Protection Act
EU	European Union
EUTR	European Timber Regulation
GCC	Gulf Co-operation Council
GSO	GCC Standardization Organization
GPSD	General Product Safety Directive
IS	Indian Standard
ISI	Indian Standards Institution
ILAC	International Laboratoy Accreditation Co-operation
ISO	International Organization for Standardization
MLA	Multilateral Recognition Arrangement
MRA	Mutual Recognition Arrangement
NABCB	National Accreditation Board for Certification Bodies
NABL	National Accreditation Board for Testing and Calibration Laboratories
REACH	Registration, Evaluation, Authorization and Restriction of Chemicals
SASO	Saudi Standards, Metrology and Quality Organization
QCI	Quality Council of India
QCO	Quality Control Order

BACKGROUND

Uttar Pradesh, the Indian state with a geographical expanse of 2, 40,928 sq.km and a population of 204.2 million people is home to a great diversity in all facets of life. With diverse community traditions and economic pursuits, Uttar Pradesh is also endowed with a beautiful diversity of crafts and industries spread across small towns and districts known for the uniqueness of these products.

To encourage such indigenous products and crafts, the Uttar Pradesh government has initiated the One District One Product (ODOP) Scheme with a vision to create product-specific traditional industrial hubs across 75 districts of Uttar Pradesh that will also promote traditional industries that are synonymous with the respective districts of the state. Although many of the district-specific industries are more of a commonplace but there are several products that are unique to those regions.

Many of these products are GI-tagged; a geographical indication (GI) is a sign used on products that have a specific geographical origin and possess qualities or a reputation that are due to that origin. This becomes a testimony to the original creativity and exclusiveness of some of the finest products produced in the various districts of Uttar Pradesh.

To help in achieving the objectives of the ODOP initiative, the ODOP Cell, Government of Uttar Pradesh, engaged Quality Council of India (QCI) to develop 'Compendiums of relevant Regulations and Standards' for the products under this initiative. The compendiums have been developed by the Quality Council of India with assistance of the sectoral and product experts. Field visits and virtual interactions with ODOP manufacturers were also undertaken to assess the level of their understanding on relevant standards and certifications. The ODOP Cell, Government of Uttar Pradesh and the EY team at the ODOP Cell also provided continuous support in the development of these compendiums.

The key objective of developing the compendiums is to provide a ready reference of various regulations & standards for the existing ODOP manufacturers and producers and also for the aspiring entrepreneurs.

This part of the compendium brings relevant regulations and standards pertaining to the Musical Instruments. Uttar Pradesh has a rich cultural heritage and was historically an important centre for innovation on music. The districts of Amroha and Pilibhit are renowned for their Dholaks and Bamboo Bansuri, respectively and are home to many units involved in their production. Further, Pilibhit is the only district in the country known for the production of bansuri.

DHOLAK, AMROHA

1. INTRODUCTION

Dholak is a two-headed hand drum, a folk percussion instrument. It is widely used in qawwali, kirtan and bhangra. Amroha district of Uttar Pradesh is a major centre for manufacture of Dholaks. There are numerous small-scale manufacturing units producing Dholak and other percussion instruments. They use the wood from Mango and Sheesham trees to carve out the multiple sized and shaped hollow blocks, which are later fitted with animal skin, mostly goatskin, to create the instrument.

A Dholak, may have traditional cotton rope lacing, screw-turnbuckle tensioning or both combined: in the first case, steel rings are used for tuning or pegs are twisted inside the laces. The smaller surface of the Dholak is made of goatskin for sharp notes and the bigger surface is made of buffalo skin for low pitches, which allows a combination of bass and treble with rhythmic high and low pitches. The shell is sometimes made from sheesham wood but cheaper dholaks may be made from any wood, such as mango.

2. RAW MATERIALS

To make a Dholak, the raw materials used are:

- Barrel shaped wood
- Dholak socket and hook
- Dholak head
- Bamboo stick
- Leather
- Ropes
- Metal rings

Figure 1: Hollowed out dholak frames

Dholak masala: It is applied mostly on the right head to lower the resonance frequency.

3. MANUFACTURING PROCESS

3.1. Cutting the wood

The seasoned wood is cut and given a barrel shape with a greater diameter at one side leaving both ends open and hollow. It is polished to a nice glossy brunet tone or any other colour desired by the buyers. The open ends are filed and shaped to fix the heads later.

3.2. Tensioning

In Screw-turn buckle tensioning: The barrel shaped wooden piece is pierced with holes evenly on both sides. The holes are made circling the openings, inches away from them. Dholak sockets are then fixed to the holes made. These sockets receive the dholak hooks which are used to secure the dholak heads in place.

Figure 2: Dholak hooks and sockets

3.3. Mounting the leather

- The processed skin or leather is soaked in water. Strips of bamboo stick is curled to the diameters of barrel shaped wooden openings (the body part of Dholak) and tied. The leather is cut to the sizes of rings made and tucked-in with help of a thin wooden stick.
- The rings wrapped with leather sheet are called Dholak heads and used to cover the wooden piece (the body part of Dholak) on both sides. The larger end of the dholak receives the Dholak head with the dholak masala (a mixture of fine sand, grease and iron fillings). This is installed with the masala facing the inside of the dholak and on the obverse side, with only a faint impression visible through the leather. The small end receives the dholak head without the masala.

Figure 3: Mounting the leather

Figure 4: Acoustic treatment

- The head is locked with help of Dholak hooks or conventional cotton rope lacings with steel rings. The hooks help to firmly fix the head and is screwed in the

- respective sockets. The heads are tuned by tightening the bolts or the hooks. It requires expertise tune the instruments and is usually carried out by the master craftsman.

4. GENERAL QUALITY TESTS

Following are a few visual quality parameters/checks:

- There should be no cracks on the Dholak
- Finish and design should be smooth
- The wood used should not be swollen
- The dholak masala should be applied uniformly and should not be in patches.

5. REGULATIONS

The regulations may be region and/or country specific and can be used in the overseas markets to protect the health and safety of consumers or be a barrier to trade, for e.g., REACH Regulations, European Union (EU), CCC mark, China, etc. Regulations (domestic) and international (some key regions/countries) are provided below. However, for other region/country specific information regarding exports may be obtained from their respective trade websites. Additional information may also be obtained from the following websites:

- Federation of Indian Export Organisations (https://www.fieo.org/)
- Directorate General of Foreign Trade (https://www.dgft.gov.in/CP/)
- Export Promotion Councils (https://commerce.gov.in/useful-links/export-promotion-councils/)

India has entered into 'Trade Agreements' including 'Free Trade Agreements' with several countries like Japan, Sri Lanka and others to facilitate mutual trade. Information regarding the FTAs may be obtained from Department of Commerce (https://commerce.gov.in/international-trade/trade-agreements/).

5.1. Domestic

S. No.	Country	Regulation	Regulatory Body	Remarks
1	India	NA	NA	NA

5.2. International

S. No.	Region/ Country	Regulation	Regulatory Body	Remarks
1	European Union	a. Registration, Evaluation, Authorization and Restriction of Chemicals (REACH) Regulations b. European Union Timber Regulation (EUTR) c. General Product Safety Directive (GPSD)	a. European Commission b. European Chemicals Agency (ECHA)	a. Legality of timber b. Traceability c. Procedural framework d. Marking and labelling

S. No.	Region/ Country	Regulation	Regulatory Body	Remarks
2	United States of America	a. Animal and Plant Health Inspection Service (APHIS) b. California Proposition 65, Act of 1986	a. Consumer Product Safety Commission (CPSC)	a. Nature of Timber (endangered or not) b. The genus and species of tree c. Quantity of regulated material
3	Australia	a. Bio Security Import Conditions System (BICON) b. Australian Competition and Consumer Commission Act 2010	a. Australian Competition and Consumer Commission (ACCC)	a. Legality of logs b. Quality of product c. Pest free product

6. VOLUNTARY STANDARDS

Domestic Standards			International Standards		
Stage	**Code**	**Aspects Covered**	**Region/ Country**	**Standard**	**Aspects Covered**
General (Refer Table 1)	**IS 707: 2011 (Reaffirmed Year: 2021)**	Timber technology and utilization of wood, bamboo and cane - Glossary of terms	European Union	**ISO 15115: 2019**	Leather -vocabulary
	IS 1640: 2007 (Reaffirmed Year: 2017)	Glossary of terms relating to hides, skins and leather	Middle east	**ISO 24294: 2021**	Timber-Round and sawn timber - Vocabulary
	IS 8171: 1992 (Reaffirmed Year: 2018)	Glossary of terms relating to polishes and related materials)	Australia	**AS/NZS 1080.3: 2000**	Vocabulary of timber
Raw Materials (Refer Table 2.a & 2.b)	**IS 8292: 1992 (Reaffirmed Year: 2021)**	Evaluation of working quality of timber under different wood working operations- Method of test	USA	**ISO 22157: 2019**	Bamboo structures — Determination of physical and mechanical properties of bamboo culms — Test methods
	IS 6874: 2008 (Reaffirmed Year: 2019)	Method of tests for bamboo	European Union	**ISO/TC 218**	Timber specification
	-	-	European Union	**ISO 22244: 2020**	Leather — Raw hides — Guidelines for preservation of hides
	-	-	USA	**ASTM D143**	Timber testing
	-	-	USA	**ASTM D4761**	Mechanical properties of wood based materials
	-	-	Middle east	**GSO 2440: 2014**	Requirements for leather
Processing (Refer Table 3)	**IS 6874: 2008 (Reaffirmed Year: 2021)**	Preservation of Timber - Code of Practice	-	-	-
	IS 1902: 2006 (Reaffirmed Year: 2021)	Preservation of bamboo (non-structural)	-	-	-

Domestic Standards			International Standards		
Stage	**Code**	**Aspects Covered**	**Region/ Country**	**Standard**	**Aspects Covered**
Finishing (Refer Table 4.a & 4.b)	**IS 582: 1970 (Reaffirmed Year: 2019)**	Chemical testing of leather, methods of sampling	European union	**ISO/TC 120**	Leather testing
	IS 5914: 1970 (Reaffirmed Year: 2019)	Methods of physical testing of leather	USA	**ASTM D 1610**	Leather testing
	-	-	Australia	**AS ISO 20137: 2017**	Chemical test of leather
	-	-	Middle East	**GSO ISO 19070: 2021**	Chemical test of leather

6.1. Table 1: (Domestic & International General Standards)

Stage	Code	Scope
General	**IS 707: 2011 (Reaffirmed Year: 2021)** - Glossary of terms	▪ Detailing of bamboo and various types of joints ▪ Requirements and specification
	IS 1640: 2007 (Reaffirmed Year: 2017) - Glossary of terms relating to hides, skins and leather	▪ Details of hides, skin and leather
	IS 8171: 1992 (Reaffirmed Year: 2018) - Glossary of terms	▪ Details about polishes and related material
	ISO 15115: 2019 - Leather - vocabulary	▪ Details of leather product
	ISO 24294: 2021 - Timber-Round and sawn timber - Vocabulary	▪ Details of various types of timber

6.2. Table 2.a: (Domestic Standards for Raw Materials)

Stage	Code	Scope
Raw Material	**IS 8292: 1992 (Reaffirmed Year: 2021)** - Evaluation of quality of timber	▪ Methods to test various types of timber. ▪ General safety provisions
	IS 6874: 2008 (Reaffirmed Year: 2019) - Method of tests for bamboo	▪ Physical properties of bamboo ▪ Mechanical properties of bamboo

6.3. Table 2.b: (International Standards for Raw Materials)

Stage	Code	Scope
Raw Material	**ISO 22157: 2019** - Bamboo structures — Determination of physical and mechanical properties of bamboo culms — Test methods	▪ Test methods for bamboo culms ▪ Physical and mechanical properties of bamboo culms
	ISO/TC 218 - Timber	▪ Standardization of processed timber used in all applications ▪ Specifications and test methods
	ISO 22244: 2020 - Leather — Raw hides — Guidelines for preservation of hides	▪ Guidelines for preservation of hides

Stage	Code	Scope
	ASTM D143 - Test methods for timber	▪ Determination of strength of wood ▪ Physical and mechanical properties of wood
	ASTM D4761 - Test method for wood-based material	▪ Mechanical properties of wood-based material ▪ Tensile, bending strength, axial strength
	AS/NZS 1080.3: 2000 - Timber-Method of test density	▪ It covers density at the moisture content at the time of test (nominal density), oven dry density and conventional density (basic density).
	GSO 2440: 2014 - General requirement for leather	▪ Specification and basic requirement for leather products

6.4. Table 3: (Domestic Standards for Processing)

Stage	Code	Scope
Processing	**IS 6874: 2008 (Reaffirmed Year: 2021)** - Preservation of timber	▪ Methods to preserve timber
	IS 1902: 2006 (Reaffirmed Year: 2021) - Preservation of bamboo (non-structural)	▪ Methods to preserve bamboo ▪ Recommendations of methods of treatment and preservatives

6.5. Table 4.a: (Domestic Standards for Finishing)

Stage	Code	Scope
Finishing	**IS 582: 1970 (Reaffirmed Year: 2019)** - Methods of chemical testing of leather	▪ Methods and chemical tests for leather
	IS 5914: 1970 (Reaffirmed Year: 2019) - Methods of physical testing of leather	▪ Methods and physical tests for leather

6.6. Table 4.b: (International Standards for Finishing)

Stage	Code	Scope
Finishing	**ISO/TC 120** - Leather	▪ Standardization in the field of: ▪ Raw hides and skins including pickled pelts; ▪ Tanned hides and skins and finished leather; ▪ Leather products (including methods of test for leather products).
	ASTM D 1610 - Standard practice for conditioning leather	▪ This practice covers the conditioning of all units and specimens of leather and leather products prior to testing and the conditions under which they should be tested. ▪ This practice does not apply to wet blue.
	AS ISO 20137: 2017 - Leather	▪ Guidelines for testing critical chemicals in leather
	GSO ISO 19070: 2021 - Leather	▪ Chemical determination of N-Methyl-2-Pyrrolidone (NMP) in leather

Note: The voluntary standards may become mandatory if the regulators of a region/ country notify the standards as a part of their regulatory requirement.

REFERENCES

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3. Domestic Standards of Dholak and raw materials: https://standardsbis.bsbedge.com/
4. ISO Standards: https://www.iso.org/obp/ui#search
5. EU : https://www.en-standard.eu/
6. USA : https://www.en-standard.eu/astm-standards/, https://www.astm.org/
7. AUS : https://www.standards.org.au/
8. GULF: https://www.gso.org.sa/store/
9. NABL accredited laboratories : https://nabl-india.org/, https://parakh.ncog.gov.in

BAMBOO FLUTES, PILIBHIT

1. INTRODUCTION

A flute is a musical instrument that produces sound when a stream of air is directed against the edge of a hole, causing the air within the body of the instrument to vibrate. It is majorly produced in Pilibhit. The district is famous for its flutes nationally as well as internationally. Pilibhit is the only district in the country known for production of Bansuri. Most flutes are tubular, but some are globular or other shapes. Some flutes are played by blowing air into a mouthpiece, which directs the air against the edge of a hole elsewhere in the flute. These instruments, known as whistle flutes, include the tubular recorder and the globular ocarina. Other flutes are played by blowing air directly against the edge of the hole.

Flute is made from a tube with holes that are covered by the fingers or keys, held vertically or horizontally so that the player's breath strikes a narrow edge. The raw material used to make flute (also known as Bansuri) comes from northeast, chiefly Silchar, Assam. There are two varieties of bansuri: transverse and fipple. The fipple flute is usually played in folk music and is held at the lips like a whistle. The transverse variety is preferred in Indian classical music as it enables superior control, variations and embellishments. Traditionally, flutes came with six holes, however seventh hole is added to improve flexibility. Bansuri or flute comes in many sizes and range between 10 inches to almost one meter. Smaller the flute, higher is the pitch of sound and as the length increases the sound becomes deeper and richer.

2. RAW MATERIALS

To make a flute, the raw materials used are:

- Good quality bamboo (Favoured bamboo variety are “Dolo” and “Muli”)
- Metals (silver, gold etc.) or wood as required
- Cork
- Decorative materials for customization
- Oil

3. MANUFACTURING PROCESS

3.1. Selecting the bamboo

The internal diameter of the bamboo is measured with a vernier calliper. Both the length and diameter ultimately decide the quality of sound. Wider the diameter and longer the length, lower the pitch of the sound.

3.2. Cutting the bamboo

The type of flute and the pitch required, decides the length. The bamboo is then measured with a measuring tape and cut to size.

3.3. Readying the reed

The inside of the bamboo is thoroughly cleaned and the knots are removed by inserting a rod of a lesser diameter through it. Then outer body and the inner areas of bamboo is smoothened with sand paper.

3.4. Making holes

A straight line is marked on the bamboo with a pencil. Usually, 7 holes are marked with a pencil. The holes are then burnt into the bamboo using hot metal rods of various diameters.

3.5. Fitting the cork

A cork of suitable thickness is fit inside the bamboo with the help of a rod. The cork is placed just above the blowing hole to prevent air from escaping.

3.6. Adjusting the length of bamboo

To obtain the right pitch, length of the bamboo is readjusted by cutting extra reed. The rest of the holes are made based on the length of the bamboo which determines the ratio that defines the position of each hole.

3.7. Tuning the flute

Although it is true that a flute is not tuned, but it may require a fine sense of pitch and precision. For example, if a particular note sounds lower, that particular hole is widened with a file until it sounds correct. However, it is almost impossible to lower a note if it is higher as the hole cannot be narrowed. One may try to fit a patch in such cases.

3.8. Finishing

Finally, the flute is then washed to remove dust, saw dust and unwanted particles. The washed and cleaned flute is then stored overnight in a cylindrical container having almond oil and antiseptic ginger oil.

4. GENERAL QUALITY TESTS

Following are a few visual quality parameters/checks:

- The carrying case should be in good condition. Scratches, dents, or any sticky residues are to be checked. The case should have a firm lock system that does not open up during any turbulence.
- The instrument should be free from cracks and scratches as they can affect the quality of music while playing.

5. REGULATIONS

The regulations may be region and/or country specific and can be used in the overseas markets to protect the health and safety of consumers or be a barrier to trade, for e.g., REACH Regulations, European Union (EU), CCC mark, China, etc. Regulations (domestic) and international (some key regions/countries) are provided below. However, for other region/country specific information regarding exports may be obtained from their respective trade websites. Additional information may also be obtained from the following websites:

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- Directorate General of Foreign Trade (https://www.dgft.gov.in/CP/)
- Export Promotion Councils (https://commerce.gov.in/useful-links/export-promotion-councils/)

India has entered into 'Trade Agreements' including 'Free Trade Agreements' with several countries like Japan, Sri Lanka and others to facilitate mutual trade. Information regarding the FTAs may be obtained from Department of Commerce (https://commerce.gov.in/international-trade/trade-agreements/).

5.1. Domestic

S. No.	Country	Regulation	Regulatory Body	Remarks
1	India	NA	NA	NA

5.2. International

S. No.	Region/ Country	Regulation	Regulatory Body	Remarks
1	European Union	**a.** Registration, evaluation, authorization and restriction of chemicals (REACH) Regulations. **b.** European Union Timber Regulation (EUTR) **c.** General Product Safety Directive (GPSD)	**a.** European Commission **b.** European Chemicals Agency (ECHA)	**a.** Legality of timber **b.** Traceability **c.** Procedural framework **d.** Marking and labelling
2	United States of America	**a.** Animal and Plant Health Inspection Service (APHIS) Regulations **b.** California Proposition 65, Act of 1986	**a.** Consumer Oroduct Safety Commission (CPSC)	**a.** Nature of Timber (endangered or not) **b.** The genus and species of tree **c.** Quantity of regulated material
3	Australia	**a.** Bio Security Import Conditions System (BICON) **b.** Australian Competition and Consumer Commission Act, 2010	**a.** Australian Competition and Consumer Commission Act (ACCC)	**a.** Legality of logs **b.** Quality of product **c.** Pest free product

6. VOLUNTARY STANDARDS

Domestic Standards			International Standards		
Stage	**Code**	**Aspects Covered**	**Region/ Country**	**Standard**	**Aspects Covered**
General (Refer Table 1)	**IS 707: 2011 (Reaffirmed Year: 2021)**	Timber technology and utilization of wood, bamboo and cane - Glossary of terms	Australia	**ISO 21625: 2020**	Vocabulary related to bamboo and bamboo products

Domestic Standards			International Standards		
Stage	**Code**	**Aspects Covered**	**Region/ Country**	**Standard**	**Aspects Covered**
	IS 6329: 2000 (Reaffirmed Year: 2021)	Code of practice for fire safety of industrial buildings - Saw mills and wood works	-	**ISO 633: 2019**	Cork- Vocabulary
Raw Materials (Refer Table 2.a & 2.b)	**IS 1902: 2006 (Reaffirmed Year: 2021)**	Preservation of bamboo and cane for non-structural purposes - Code of practice	USA	**ISO 1997: 2018**	Granulated Cork and Cork Powder: Classification, packing
	IS 6874: 2008 (Reaffirmed Year: 2019)	Methods of test for bamboo	-	-	-
	IS/ISO 2030: 2018	Granulated cork - Size analysis by mechanical sieving	-	-	-
	IS/ISO 2031: 2015	Granulated cork - Determination of apparent bulk density	-	-	-
	IS/ISO 2067: 2019	Granulated cork: Sampling	-	-	-
	IS/ISO 2190: 2016	Granulated cork-Determination of moisture content	-	-	-

6.1. Table 1: (Domestic & International General Standards)

Stage	Code	Scope
General	**IS 6329: 2000 (Reaffirmed Year: 2021)** - Code of practice for fire safety of industrial buildings - Saw mills and wood works	▪ Various measures to prevent fire in factories
	IS 707: 2011 (Reaffirmed Year: 2021) - Timber technology and utilization of wood, bamboo and cane - Glossary of terms	▪ Detailing of bamboo and various types of joints ▪ Requirements and specification
	ISO 633: 2019 - Cork-Vocabulary	▪ Vocabulary for all types of cork
	ISO 21625: 2020 - Vocabulary related to bamboo and bamboo products	▪ Glossary of terms

6.2. Table 2.a: (Domestic Standards for Raw Materials)

Stage	Code	Scope
Raw Material	**IS 1902: 2006 (Reaffirmed Year: 2021)** - Preservation of Bamboo (non-structural)	▪ Methods to preserve bamboo ▪ Recommendations of methods of treatment and preservatives
	IS 6874: 2008 (Reaffirmed Year: 2019) - Method of tests for bamboo	▪ Physical properties of bamboo ▪ Mechanical properties of bamboo
	IS/ISO 2030: 2018 - Granulated cork — Size analysis by mechanical sieving	▪ Methods of analysing cork size
	IS/ISO 2031: 2015 - Granulated cork — Determination of apparent bulk density	▪ Methods to determine the bulk density of granulated cork
	IS/ISO 2067: 2019 - Granulated cork — Sampling	▪ Sampling methods

Stage	Code	Scope
	IS/ISO 2190 - Granulated cork-Determination of moisture content	▪ Methods to determine the moisture content in granulated cork

6.3. Table 2.b: (International Standards for Raw Materials)

Stage	Code	Scope
Raw Material	**ISO 1997: 2018** - Granulated Cork and Cork Powder: Classification, packing	▪ Classification of cork ▪ Properties of cork ▪ Packaging of cork

Note: The voluntary standards may become mandatory if the regulators of a region/ country notify the standards as a part of their regulatory requirement.

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4. ISO Standards: https://www.iso.org/obp/ui#search
5. EU: https://www.en-standard.eu/
6. USA: https://www.en-standard.eu/astm-standards/, https://www.astm.org/
7. AUS: https://www.standards.org.au/
8. GULF: https://www.gso.org.sa/store /
9. NABL accredited laboratories: https://nabl-india.org/, https://parakh.ncog.gov.in

APPENDIX I: TESTING AND CERTIFICATION

Uttar Pradesh is home to a large number of indigenous products and manufacturing units associated with these unique products. The crafts and artisans have both withstood the vagaries of time and have flourished under the patronage provided in the form of various schemes to promote these products and bring about innovation in the industry by up skilling the artisans as well as providing them assistance in tapping new markets and becoming competitive in the modern market by utilizing machinery developed locally to augment their efforts.

With the advent of machinery and technology there has been a gradual move towards standardization to improve both the product life cycle as well as the quality of materials used. Two important aspects in standardization of products and processes are:

- Testing (Product/Material testing)
- Certification (Product Certification or Systems Certification)

1. PRODUCT TESTING

What is product testing?

Testing is the determination of one or more of a product's characteristics and is usually performed by a laboratory. In other words, product testing is a process to measure its performance, safety, quality, and compliance with established standards.

Quality testing is involved at each step of the production process and often begins with the testing of raw materials, moving through the testing of in-process products to the finished product.

Testing is an essential part of developing a quality product. Any product that is produced is expected to meet certain requirements and specifications to ensure standardization and consistency.

Why is testing of products required?

Testing is an essential part of developing a quality product. Product testing is required to assure quality, performance and safety of the product. Product testing is also required to comply with the regulatory requirements and standards, wherever prescribed.

Testing at the various stages of production helps to ensure that the products meet the defined specifications and gain acceptance globally.

Why should you get the product testing done from an approved/ accredited laboratory?

It is always recommended to get the tests done at a Laboratory that is approved /accredited by a competent authority. Accreditation indicates a formal recognition by an independent body, generally known as an accreditation body, which a certification body operates according to international standards.

Accreditation is an international network which is managed by the International Accreditation Forum (IAF), in the fields of management systems, products, services, personnel and other similar programmes of conformity assessment, and the International Laboratory Accreditation Cooperation (ILAC), in the field of laboratory and inspection accreditation. ILAC and IAF are international associations of accreditation bodies and they facilitate world trade through mutual recognition and also enhances the acceptance of products and services across national borders. Details on these international bodies and their MRA framework are available on www.iaf.nu and www.ilac.org, respectively.

In India, the National Accreditation Board for Testing and Calibration Laboratories (NABL), a constituent Board of Quality Council of India (QCI) has been established as the National Accreditation Body for accreditation of such laboratory. NABL is Mutual Recognition Arrangements (MRA) signatory to ILAC as well as APAC for the accreditation of Testing and Calibration Laboratories (ISO/IEC 17025), Medical Testing Laboratories (ISO 15189), Proficiency Testing Providers (PTP) (ISO/IEC 17043) and Reference materials producers (RMP).

Such MRA reduces technical barrier to trade and facilitates acceptance of test/ calibration results between countries which MRA partners represent.

Further, for certain products like food, pharmaceuticals etc., there are regulatory requirements (domestic as well as region or country specific (if exporting)) for testing of the raw materials and finished goods which are mandatory to be followed. For such products, various regulations and the standards making bodies like Bureau of Indian Standards (BIS) also notify specific Laboratories, from time to time. For e.g. BIS has recognised several laboratories for the tests prescribed under the Indian Standards formulated by them. Similarly, other regions or countries may have specific regulations that require prescribed tests, for e.g. REACH regulation of the European Union, specifies requirements to for the protection of human health and the environment from the risks that can be posed by chemicals.

Where can you get the list of accredited/approved laboratories?

List of the testing laboratories for the relevant tests may be obtained from the following URLs:

- **National Accreditation Board for Testing and Calibration Laboratories (NABL)**: https://nabl-Indiaorg/
- **Bureau of Indian Standards (BIS)**: https://bis.gov.in/

2. CERTIFICATION

Certification is a provision by an independent body of written assurance (a certificate) that the product, service or system in question meets specific requirements.

Broadly, certifications may be categorised as:

a. Management Systems Certification

These certifications help organizations improve their performance by specifying repeatable steps that organizations consciously implement to achieve their goals and objectives, and to create an organizational culture that reflexively engages in a continuous cycle of self-evaluation, correction and improvement of operations and processes through heightened employee awareness and management leadership and commitment.

The benefits of an effective management system to an organization include:

- More efficient use of resources and improved financial performance
- Improved risk management and protection of people and the environment
- Increased capability to deliver consistent and improved services and products, thereby increasing value to customers and all other stakeholders

Examples of Systems Certification include ISO 9001 QMS, ISO 22000 FSMS, etc.

b. Product Certification

Product certification is the process of certifying that a specific product fulfils requirements prescribed in contracts, regulations or specifications and has passed all necessary quality assurance and performance tests. Product certification is often required in industries and marketplaces where product failure could bring serious negative consequences to the health and safety of people and environment.

The benefits of an effective management system to an organization include:

- Product certification gives product users the confidence they need to purchase the desired products.
- If a product is certified according to applicable national or international laws, customers know that that product will function correctly and won't cause any harm. Examples of product certification include CE marking, ISI mark, Hallmark, etc.

Are all certifications mandatory?

Mostly, the certifications are voluntary in nature. However, for a number of products, compliance to specific Standards may be made compulsory, for instance, compliance to Indian Standards for a number of products, has been made compulsory by the Central Government under various considerations viz. public interest, protection of human, animal or plant health, safety of environment, prevention of unfair trade practices and national security. For such products, the Central Government directs mandatory use of Standard Mark under a Licence or Certificate of Conformity (CoC) from BIS through issuance of Quality Control Orders (QCOs). Similarly, other regions or countries may have specific regulations that require prescribed certifications, for e.g. USFDA prescribes specific requirements for Food, Drugs and Cosmetic products.

Why should you get the Certification from an accredited Certification Body?

It is always recommended to obtain the certification from accredited Certification Bodies or the ones that are prescribed by the relevant regulatory authorities. Accreditation indicates a formal recognition by an independent body, generally known as an accreditation body, which a certification body operates according to international standards.

Accreditation is an international network which is managed by the International Accreditation Forum (IAF), in the fields of management systems, products, services, personnel and other similar programmes of conformity assessment, and the International Laboratory Accreditation Cooperation (ILAC), in the field of laboratory and inspection accreditation. ILAC and IAF are international associations of accreditation bodies and they facilitate world trade through mutual recognition and also enhances the acceptance of products and services across national borders. Details on these international bodies and their MRA framework are available on www.iaf.nu and www.ilac.org, respectively.

In India, the National Accreditation Board for Certification Bodies (NABCB), a constituent board of Quality Council of India (QCI) has been established as the National Accreditation Body for accreditation of such Certification and Inspection Bodies. The National Accreditation Board for Certification Bodies provides accreditation to Certification and Inspection Bodies based on assessment of their competence as per the Board's criteria and in accordance with International Standards and Guidelines.

NABCB is internationally recognized and represents the interests of the Indian industry at international forums through membership and active participation with the objective of becoming a signatory to international Multilateral / Mutual Recognition Arrangements (MLA / MRA).

Where can you get the list of accredited/approved laboratories?

List of the NABCB accredited Certification Bodies (CBs) and Inspection Bodies (IBs) for relevant certifications may be obtained from the following URL:

National Accreditation Board for Certification Bodies (NABCB):
http://nabcb.qci.org.in/

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ODOP Cell, Directorate of Industries, Niryat Bhawan, First Floor, 8 Cantt Road, Qaiserbagh, Lucknow – 226001